

The First Law on the Net: The Stress Tensor, the Wedge Modular Hamiltonian, and the Relative Entropy of Coherent Excitations

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****Abstract.**** This note opens the interaction era at the point the era analysis fixed: the free-theory brick on which the entropic-gravity program stands. First, the stress tensor is **derived**, not chosen: given the series' symplectic normalization $\sigma(F, G) = 2\pi \int F' G$, the requirement that the energy density generate chiral translations under the Poisson structure forces $T(r) = \frac{1}{4\pi} :j(r)^2:$ uniquely, and the wedge modular Hamiltonian $K_{\mathcal{N}} = 2\pi \int_0^\infty r T(r) dr$ is then the second-quantized generator of the series' verified modular flow — with the Bisognano-Wichmann orientation explicit: Δ^{it} implements the dilations run backwards, $K = 2\pi d\Gamma(-D)$. The central result is the ****exact first law**** for coherent excitations supported in the wedge: the Araki relative entropy of the coherent state $W(F)\Omega$ with respect to the wedge algebra equals the classical dilation energy of its wave,

$$S(\omega_F \parallel \omega)|_{\mathcal{N}} = \Delta \langle K_{\mathcal{N}} \rangle = \frac{1}{2} \int_0^\infty r \ell(r)^2 dr, \quad \ell = 2\pi F',$$

exactly, not merely to first order. The proof runs through the unitary identity $S(\omega \circ \text{Ad } U^* \parallel \omega) = \langle U\Omega, (-\ln \Delta_\Omega)U\Omega \rangle$ for $U \in \mathcal{N}$, the verified modular structure, and the Weyl displacement, which reduces everything to a one-particle dilation energy. Verified by two independent computation routes — the spectral form $2\pi \langle KF, (-D)KF \rangle$ in Fourier-Mellin variables against the position-space energy integral — agreeing at 1.2×10^{-6} , 5.9×10^{-6} , 1.2×10^{-6} across three wave packets, with the dilation-weight scaling under outward translation bit-exact. Positivity of relative entropy is realized classically, manifest in the formula. With the interior modular Hamiltonians of the net paper and this law, the equation $\delta S = \delta \langle K \rangle$ — the engine of the Jacobson/entanglement-equilibrium derivation of gravity — is now a theorem **inside** the model, in objects the model owns. Boundary-crossing waves, diamonds, the measurement gate, and the covariance audit are scoped as the era's named next steps.

1. Step one of the era

The era analysis concluded: gravity's entropic foundation — the first law relating entropy variations to modular energy — requires a stress tensor and state perturbations, both available in the free theory; interactions enter only afterwards. This note delivers that foundation as an internal theorem with verified constants. The inputs are all series-owned: the local net with its wedge modular structure (interior-net paper; localization and geometrization papers), Araki's relative entropy — finite and regulator-free on the type III₁ algebras the series twice derived — and the coherent excitations of the free field.

2. The stress tensor, derived

The series fixed the current's symplectic form as $\sigma(F, G) = 2\pi \int F' G du$. On the classical phase space this is the Poisson structure $\{\ell(r), \ell(r')\} = -2\pi \delta'(r - r')$ for the classical

current ℓ . Requiring the Hamiltonian $H = \alpha \int \ell^2 dr$ to generate chiral translations, $\{\ell, H\} = -\ell'$, gives $\{\ell(r), H\} = -4\pi\alpha \ell'(r)$, hence $\alpha = 1/4\pi$ *uniquely*:

$$T(r) = \frac{1}{4\pi} :j(r)^2:, \quad K_{\mathcal{N}} = 2\pi \int_0^\infty r T(r) dr.$$

This $K_{\mathcal{N}}$ is the second-quantized generator of exactly the wedge modular flow the series has verified — via detailed balance on the scale line and the exponential correspondence — with one orientation fact stated rather than hidden: the modular group of the right wedge implements the dilations *run backwards*, $\Delta^{it} = e^{-2\pi it d\Gamma(D)}$, so $K_{\mathcal{N}} = -\ln \Delta_{\mathcal{N}} = 2\pi d\Gamma(-D)$, positive on wedge excitations. (The verification below caught this sign before the theorem was stated; the transcript keeps the episode.)

3. The coherent family

For real $F \in C_c^\infty((0, \infty))$, the Weyl operator $W(F) = e^{ij(F)} \in \mathcal{N}$ shifts the current by a classical wave: from $[j(r), j(r')] = -2\pi i \delta'(r - r')$,

$$W(F)^* j(r) W(F) = j(r) + \ell(r), \quad \ell(r) = 2\pi F'(r),$$

so the coherent state $\omega_F = \langle W(F)\Omega, \cdot W(F)\Omega \rangle$ is the vacuum with a classical current profile ℓ riding on it. Its classical dilation energy in the wedge is $\int_0^\infty r t_{\text{cl}}(r) dr$ with $t_{\text{cl}} = \ell^2/4\pi$.

4. Theorem: the exact first law

****Theorem.**** For F supported in the wedge,

$$S(\omega_F \parallel \omega)|_{\mathcal{N}} = \Delta \langle K_{\mathcal{N}} \rangle_F = 2\pi \int_0^\infty r \frac{\ell(r)^2}{4\pi} dr = \frac{1}{2} \int_0^\infty r \ell(r)^2 dr = 2\pi^2 \int_0^\infty r F'(r)^2 dr,$$

exactly.

Proof. (i) For a unitary U in a von Neumann algebra $\mathcal{N}$ with cyclic separating Ω , Araki's relative entropy obeys $S(\omega \circ \text{Ad } U^* \parallel \omega) = \langle U\Omega, (-\ln \Delta_\Omega) U\Omega \rangle$ — verified in the type-I model by the trace computation $S(U\rho U^* \parallel \rho) = \langle -\ln \rho \rangle_{U\rho U^*} - \langle -\ln \rho \rangle_\rho$, with the commutant term cancelling. Apply it with $U = W(F) \in \mathcal{N}$. (ii) By the series' wedge modular structure, $-\ln \Delta_{\mathcal{N}} = 2\pi d\Gamma(-D)$ with D the one-particle dilation generator; on the one-particle Fourier side ($\langle KF, KG \rangle = \int_0^\infty k \hat{F}^* \hat{G} dk$), $-D$ acts as $+i(k\partial_k + 1)$. (iii) The Weyl displacement reduces the quadratic generator's expectation to its one-particle value on the shift: $\langle W(F)\Omega, 2\pi d\Gamma(-D) W(F)\Omega \rangle = 2\pi \langle KF, (-D) KF \rangle$, the linear terms vanishing in the vacuum. (iv) A Parseval-type identity converts the spectral form to the position-space energy integral above. ■

The law is exact on the coherent family — no linearization — and Araki positivity is realized classically: $S = \frac{1}{2} \int r \ell^2 \geq 0$, manifestly.

5. Verification

The spectral and position sides were computed by independent routes — the Fourier-Mellin form $2\pi \int_0^\infty k \hat{F}^*(i)(k\hat{F}' + \hat{F}) dk$ with the k -derivative taken as an exact second transform, against $2\pi^2 \int r F'^2 dr$ — for three wedge-supported wave packets: relative deviations $1.15 \times 10^{-6}, 5.90 \times 10^{-6}, 1.15 \times 10^{-6}$, imaginary parts at quadrature noise. The dilation-weight structure was tested separately: translating a packet outward by equal steps changed S by increments identical to all computed digits (16.169813, 16.169813) — the linear-in- r weighting of the modular Hamiltonian read off from the numbers themselves.

6. The law and the program

What this establishes for the series: $\delta S = \delta \langle K \rangle$ — the identity from which the Jacobson equation-of-state and entanglement-equilibrium derivations of the Einstein equations proceed — is now a *theorem of the model*, holding exactly on a dense family of excitations, with the K 's the interior-net paper supplied (wedges here; diamonds via the Hislop-Longo flows next) and the entropy Araki's, regulator-free on the III_1 algebras the series derived from the octonionic side. Every object in the law is series-owned. The gravitational step proper — perturbing the geometry and demanding the law hold in all local frames — is now a formulated problem rather than a program slogan; and what interactions will add is equally sharp: energy exchange between excitations, and the sector dynamics of the soft-sector paper, since radiating waves pump memory.

7. Scope

Five items. (i) The theorem covers wedge-supported waves, where $W(F) \in \mathcal{N}$ makes the unitary identity exact. Waves crossing the wedge edge are the harder case, established in the literature (Longo; Casini-Grillo-Pontello; Hollands for the chiral setting) with a known subtlety this series flags for its own future use: the stress-tensor improvement ambiguity contributes boundary terms at the entangling surface. (ii) The statement is per-generator chiral; the 3+1 wedge version follows by the assembly's generator sum and is noted, not executed. (iii) Diamonds — the Hislop-Longo modular Hamiltonians — are the next brick, and with them the entanglement-equilibrium formulation on causal diamonds. (iv) The measurement gate (the Fewster-Verch probe coupled to this net) is the era's Step 2, per the era analysis. (v) The covariance audit of the native interaction candidate — the tower's finite- μ remnants against the assembled boosts — is Step 3, and remains the era's central open risk.

References

- Companion papers: *The Interior Net: Local Algebras from the Cone and Microcausality as Output*; *The Assembly: Null Lines Through a Point, the Light-Cone Identification, and Lorentz Symmetry as Output*; *Localization on the Scale Line: The Local Current Net, the Thermal Wedge, and the Modular Seed of the Möbius Group*; *Geometrization: The Exponential Correspondence, Thermal Relative Duality, and the Shadow's Home*; *The Soft Sector: Memory as a Superselection Label and the Harmonic-Number Structure of the Zero-Mode Kernel*.

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